

Akshay Kumar Mundrathi

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EDUCATION

Oklahoma State University — Stillwater, OK

M.S. in Computer Science — GPA: 3.64

Aug 2023 – May 2025

National Institute of Technology Warangal — Warangal, India

B.Tech in Computer Science

Aug 2019 – May 2023

EXPERIENCE

Research Assistant — Well-Records Digitization Pipeline

Jul 2025 – Present

Boone Pickens School of Geology, Oklahoma State University — Stillwater, OK

- Inherited a single-document research prototype and engineered a production pipeline (Google Vision OCR, OpenCV, U-Net segmentation) that digitized 570,000+ historical Oklahoma well records, mapping 51,000+ wells across all 77 counties.
- Built a 5-pass coordinate resolver against a 4.5M-cell PostGIS database on AWS RDS and designed a self-correcting public map with an automated GitHub Actions correction pipeline.
- Presented “Automated Extraction of Well Coordinates from Oklahoma 1002A Records” as a poster at the AAPG 4th Annual Orphan, Abandoned, Idle & Marginal Wells Conference (Mar 2026).

Graduate Research & Teaching Assistant

Jan 2025 – May 2025

Oklahoma State University — Stillwater, OK

- Led discussion sections and graded coursework for Discrete Mathematics while completing graduate coursework.
- Gained hands-on experience with core AWS services (EC2, S3, IAM) and cloud lifecycle management best practices.

IT Intern

Aug 2024 – Jan 2025

Oklahoma State University Police Department — Stillwater, OK

- Delivered hardware and software support for faculty and officers, maintaining network devices, workstations, and peripherals.
- Diagnosed and resolved recurring technical issues, keeping systems updated and performant department-wide.

Data Analyst

Aug 2022 – Jun 2023

Cognizant — India

- Built Python/SQL data pipelines and Excel/macro-driven dashboards supporting business reporting for cross-functional stakeholders.
- Bridged business and technology teams across Waterfall and Agile projects, presenting analytical findings directly to senior leadership.

PROJECTS

Spiking Neural Network — Financial Time Series Classification | GitHub

- Engineered a Python pipeline using Spiking Neural Networks to classify 522 S&P 500 time-series segments into 5 trend classes, reporting 89% precision / 88% recall.
- Built a 25-input/5-output single-layer SNN with lateral inhibition, converting raw data into spike-train encodings and permutation matrices for biologically-plausible pattern recognition.

Modern Data Warehouse & ETL Pipeline (SQL Server) | GitHub

- Architected a Medallion-Architecture (Bronze/Silver/Gold) warehouse integrating 10+ heterogeneous sources, cutting manual data prep time 60% and improving pipeline efficiency 40%.
- Designed Star-Schema dimensional models and real-time Power BI dashboards, reducing reporting latency 50% for cross-functional decision-making.

Cricket Analytics — Web Scraping to Power BI | GitHub

- Automated extraction of 10,000+ match statistics via BeautifulSoup/Requests and built a Pandas-based cleaning pipeline, cutting preprocessing time 30%.
- Delivered interactive Power BI dashboards surfacing player and match trends, increasing stakeholder insight generation by 50%.

TECHNICAL SKILLS

Languages & Scripting: Python, SQL, JavaScript, Bash, PowerShell

Data & Machine Learning: Pandas, NumPy, Scikit-learn, TensorFlow, Power BI, ETL, Data Modeling

Web & Databases: React.js, Node.js, Express.js, MongoDB, SQL Server, PostgreSQL/PostGIS

Cloud & DevOps: AWS (EC2, S3, IAM, RDS), GitHub Actions, CI/CD

PUBLICATIONS & CERTIFICATIONS

- “Improved ANFIS Model Based on Confinement Optimized PSO to Predict the Air Quality of Delhi, India” — *IEEE FUZZ 2023*, Incheon, South Korea (presented in person)
- AWS Certified Developer – Associate | Google Data Analytics Professional Certificate